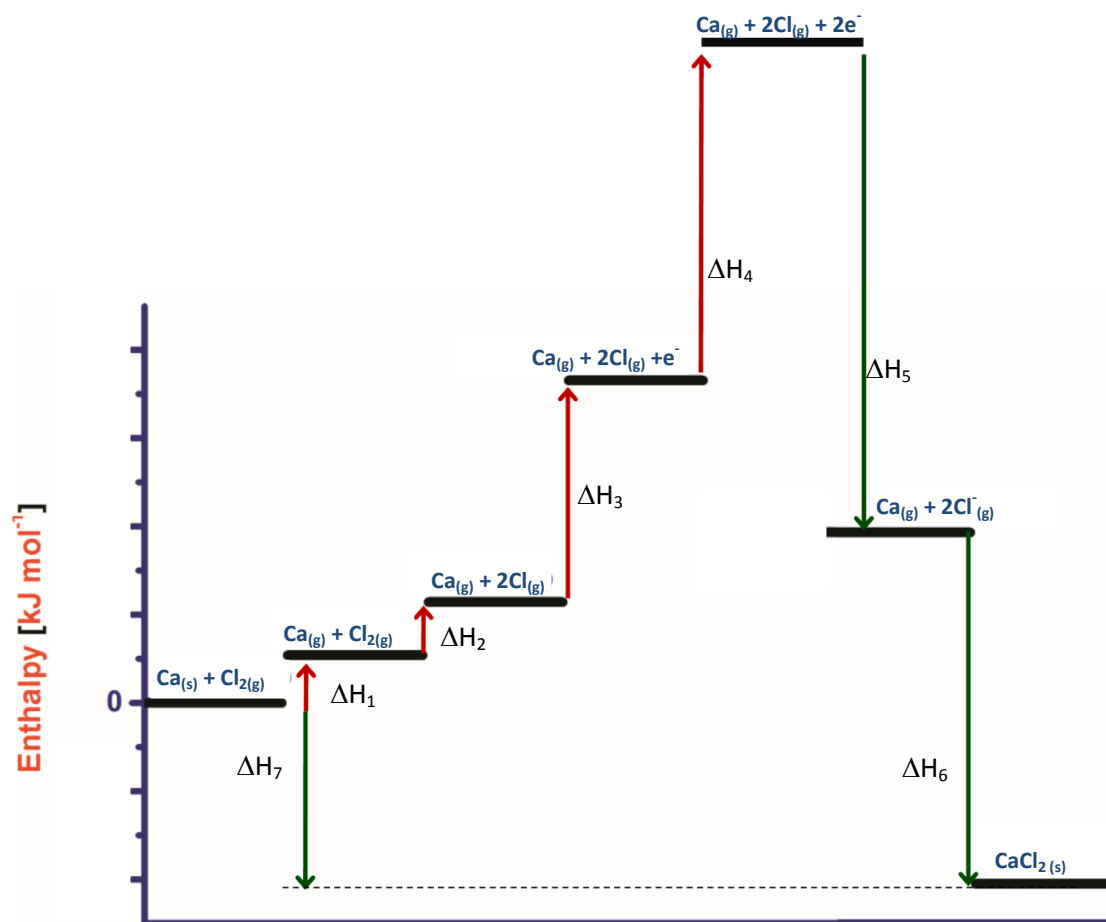


1. Represent the Born-Haber cycle for CaCl_2 . What kind of information can we obtain from this thermodynamic cycle? (10pt.)



$\Delta H_1 = \Delta H^\circ_{\text{vap}}$ = Enthalpy of vaporization for Ca;

$\Delta H_2 = \Delta H^\circ_{\text{diss}}$ = Dissociation of Cl_2 into gaseous atoms;

$\Delta H_3 = \Delta H^\circ_{\text{ion1}}$ = first ionization enthalpy of gaseous Ca;

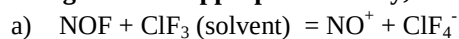
$\Delta H_4 = \Delta H^\circ_{\text{ion2}}$ = second ionization enthalpy of gaseous Ca;

$\Delta H_5 = 2 \times \Delta H^\circ_{\text{EA}}$ = 2 x electron affinity of Cl;

$\Delta H_6 = \Delta H^\circ_{\text{Cryst.}}$ = U = Lattice energy of CaCl_2 ;

$\Delta H_7 = \Delta H^\circ_{\text{f.}}$ = CaCl_2 formation energy.

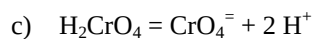
2. Using the most appropriate theory, identify acids and bases in the following reactions (5pt.):



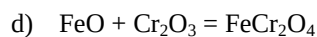
NOF is a base because it increases the concentration of the anionic species (solvent-system definition).



XeO₃ is an acid because it is removing OH⁻ species (solvent-system definition, but also Lewis definition).



H₂CrO₄ is an acid because it supplies protons (Brønsted definition).



FeO is a base and Cr₂O₃ is an acid because the first one is giving and oxygen to the second one (Lux-Flood definition).



(CH₃)₃N is a base and BF₃ is an acid (Lewis definition).

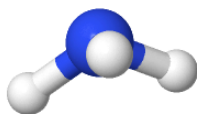
4. For each coordination number from three to six, draw the principal geometrical arrangement (or arrangements) and, if possible, make an example. (8pt.)

Coordination Number 3:

Trigonal planar, e.g. $[\text{HgI}_3]^-$.

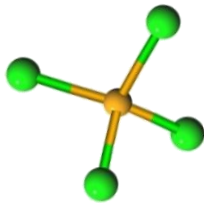


Pyramidal, e.g. SnCl_3 .

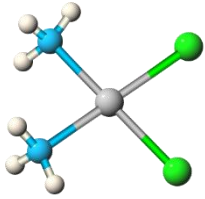


Coordination Number 4:

Tetrahedral, e.g. FeCl_4^- .

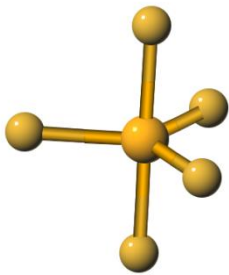


Square planar, e.g. $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$.

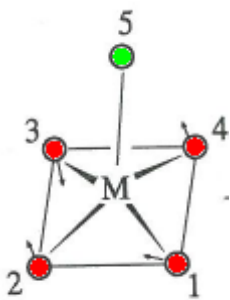


Coordination Number 5:

Trigonal bipyramid, e.g. $\text{Fe}(\text{CO})_5$

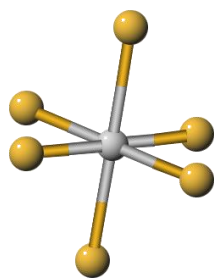


Squared pyramid, e.g. $[\text{Ni}(\text{CN})_5]^{3-}$



Coordination Number 6:

Octahedral, e.g. $[\text{TiF}_6]^{2-}$

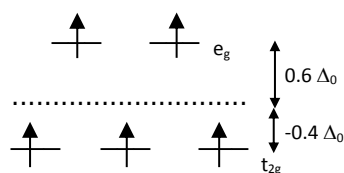


5. By using the Crystal field theory explain why Mn_3O_4 has normal spinel structure and Fe_3O_4 inverse spinel structure.

a) Mn_3O_4

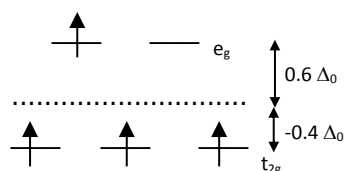
Oxygen is a weak field ligand (high spin complexes)

Mn^{2+} has 5 electrons in its $3d$ orbitals



$$\text{Crystal Field Stabilization Energy (CFSE)} = 3 \times (-0.4\Delta_0) + 2 \times (0.6\Delta_0) = 0$$

Mn^{3+} has 4 electrons in its $3d$ orbitals



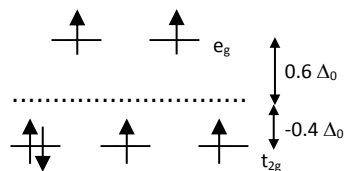
$$\text{Crystal Field Stabilization Energy (CFSE)} = 3 \times (-0.4\Delta_0) + 1 \times (0.6\Delta_0) = -0.6\Delta_0$$

The stabilization of the octahedral field is higher (in absolute value) for Mn^{3+} than for Mn^{2+} , which means that the most stable structure has most likely Mn^{3+} ions in the octahedral cavities, therefore we can predict a NORMAL spinel structure for this compound.

b) Fe_3O_4

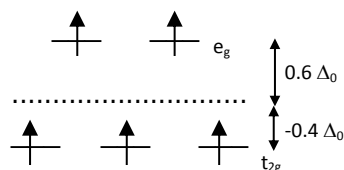
Oxygen is a weak field ligand (high spin complexes)

Fe^{2+} has 6 electrons in its 3d orbitals



$$\text{Crystal Field Stabilization Energy (CFSE)} = 4 \times (-0.4\Delta_0) + 2 \times (0.6\Delta_0) = -0.4\Delta_0$$

Fe^{3+} has 5 electrons in its 3d orbitals



$$\text{Crystal Field Stabilization Energy (CFSE)} = 3 \times (-0.4\Delta_0) + 2 \times (0.6\Delta_0) = 0$$

The stabilization of the octahedral field is higher (in absolute values) for Fe^{2+} than for Fe^{3+} , which means that the most stable structure has most likely Fe^{2+} ions in the octahedral cavities, therefore we can predict an INVERSE spinel structure for this compound.